

Word Problems & Applications

Translating · Number · Money · Mixture · Distance-Rate-Time · Percent · Geometry

MTH1W · Class — Guided Practice

Name: _____

Date: _____

Goal: Translate real-world situations into equations and solve number, money, mixture, distance, percent and geometry problems.

Warm-up (quick recall)

1. Write an expression for 'a number n increased by 7'.
2. Write an expression for 'twice a number, decreased by 5'.
3. The sum of a number and 12 is 20. Write the equation.

Examples & Practice

For each problem type, study the worked **Example**, then solve the questions that follow. Difficulty rises within each set (L1 → L3). Show your setup (let statement + equation). Answers are on the last page.

Translating Words into Equations

Example:

Write an equation: 'When 6 is added to three times a number, the result is 27.' Then solve.

Let the number be n : $3n + 6 = 27$. Subtract 6: $3n = 21$.

Divide by 3: $n = 7$. Answer: the number is 7.

4. Write an expression for 'five more than a number x '. **[L1]**

5. Write an equation for 'a number decreased by 4 equals 10', then solve. **[L1]**

6. Twice a number, increased by 9, is 31. Find the number. **[L2]**

7. When a number is multiplied by 4 and then 7 is subtracted, the result equals 3 times the number plus 2. Find the number. **[L3]**

Number Problems (Consecutive Integers, 'Think of a Number')

Example:

The sum of three consecutive integers is 48. Find them.

Let the integers be n , $n+1$, $n+2$: $n + (n+1) + (n+2) = 48$.

$3n + 3 = 48 \rightarrow 3n = 45 \rightarrow n = 15$. Answer: 15, 16, 17.

8. Two consecutive integers have a sum of 27. Find them. **[L1]**

9. The sum of three consecutive even integers is 90. Find them. **[L2]**

10. I think of a number, double it, and add 5. The result is 41. What was my number?
[L2]

11. The sum of three consecutive odd integers is 111. What is the largest? **[L3]**

Money Problems (Coins, Prices, Wages)

Example:

A part-time job pays a \$20 flat fee plus \$14 per hour. How many hours give \$104?
Let h be the hours: $14h + 20 = 104$. Subtract 20: $14h = 84$.
Divide by 14: $h = 6$. Answer: 6 hours.

12. Tickets cost \$8 each. How many tickets can be bought with \$56? **[L1]**
13. A phone plan charges \$25 per month plus \$0.10 per text. One month's bill was \$32. How many texts were sent? **[L2]**
14. Maria has \$2.35 in dimes and quarters, 13 coins in total. How many of each? (Let d = dimes.) **[L2]**

15. A worker earns \$18/h for regular hours and \$27/h for overtime. In one week she worked 40 regular hours plus some overtime and earned \$963. How many overtime hours? **[L3]**

Mixture Problems (Concentration / Blends)

Example:

A store blends coffee worth \$8/kg with coffee worth \$12/kg to make 10 kg of blend worth \$9.20/kg. How many kg of each?

Let x = kg of the \$8 coffee, so $(10 - x)$ = kg of the \$12 coffee.

$8x + 12(10 - x) = 9.20(10) \rightarrow 8x + 120 - 12x = 92 \rightarrow -4x = -28 \rightarrow x = 7$.

Answer: 7 kg of the \$8 coffee and 3 kg of the \$12 coffee.

16. A 500 mL drink is 20% juice. How many millilitres of juice does it contain? **[L1]**
17. How many litres of pure water must be added to 3 L of 40% saltwater to dilute it to 30% salt? **[L2]**

18. Nuts worth \$6/kg are mixed with nuts worth \$10/kg to make 8 kg of mix worth \$7.50/kg. How many kg of the \$6 nuts? **[L2]**
19. A chemist has 40 mL of 25% acid solution. How many millilitres of pure (100%) acid must be added to make a 40% acid solution? **[L3]**

Distance-Rate-Time

Example:

A train travels 240 km in 3 hours at constant speed. Find its speed, then the distance it covers in 5 hours at that speed.

Speed = distance $\div$ time = $240 \div 3 = 80$ km/h.

Distance in 5 h = $80 \times 5 = 400$ km.

20. A car travels at 60 km/h for 2 hours. How far does it go? **[L1]**

21. A runner covers 21 km in 1.75 hours. Find the average speed in km/h. **[L2]**
22. A plane flies 1500 km at 500 km/h. How many hours does the flight take? **[L2]**
23. Two cars leave the same point in opposite directions at 70 km/h and 90 km/h. After how many hours are they 480 km apart? **[L3]**

Percent Problems (Discount, Tax, Increase/Decrease, Work Backwards)

Example:

A jacket is marked \$80. It is discounted 25%, then 13% tax is added. Find the final price.

Discount: $80 \times 0.25 = 20$, so the sale price is 60.

Tax: $60 \times 0.13 = 7.80$, so the final price is $60 + 7.80 = \$67.80$.

24. Find 15% of \$60. **[L1]**

25. A \$45 item is on sale for 20% off. What is the sale price? **[L2]**

26. After a 13% tax is added, a meal costs \$56.50. What was the price before tax? **[L2]**

27. A population increased by 20% to reach 3600. What was the original population?
[L3]

Geometry Word Problems (Perimeter / Area / Angles as Equations)

Example:

The length of a rectangle is 5 cm more than its width. The perimeter is 46 cm. Find the dimensions.

Let w = width, so length = $w + 5$. $P = 2(w + w + 5) = 46$.

$2(2w + 5) = 46 \rightarrow 4w + 10 = 46 \rightarrow 4w = 36 \rightarrow w = 9$.

Answer: width 9 cm, length 14 cm.

28. A square has perimeter 36 cm. Find the length of one side. **[L1]**
29. Two angles are complementary (sum to 90°). One is twice the other. Find both angles. **[L2]**
30. A rectangle is 3 times as long as it is wide. Its perimeter is 64 m. Find its dimensions. **[L2]**

31. The angles of a triangle are in the ratio 1:2:3. Find each angle. **[L3]**

Answers

Warm-up: W1) $n + 7$ W2) $2n - 5$ W3) $n + 12 = 20$

4. (L1) $x + 5$
5. (L1) $n - 4 = 10$, so $n = 14$.
6. (L2) $2n + 9 = 31 \rightarrow 2n = 22 \rightarrow n = 11$.
7. (L3) $4n - 7 = 3n + 2 \rightarrow n = 9$. (Check: $4(9) - 7 = 29 = 3(9) + 2$.)
8. (L1) $n + (n+1) = 27 \rightarrow 2n + 1 = 27 \rightarrow n = 13$; integers 13 and 14.
9. (L2) $n + (n+2) + (n+4) = 90 \rightarrow 3n + 6 = 90 \rightarrow n = 28$; integers 28, 30, 32.
10. (L2) $2n + 5 = 41 \rightarrow 2n = 36 \rightarrow n = 18$.
11. (L3) $3n + 6 = 111 \rightarrow n = 35$; integers 35, 37, 39; largest is 39.
12. (L1) $8t = 56 \rightarrow t = 7$ tickets.
13. (L2) $25 + 0.10t = 32 \rightarrow 0.10t = 7 \rightarrow t = 70$ texts.
14. (L2) $0.10d + 0.25(13 - d) = 2.35 \rightarrow -0.15d = -0.90 \rightarrow d = 6$; so 6 dimes and 7 quarters.
15. (L3) $40(18) + 27x = 963 \rightarrow 720 + 27x = 963 \rightarrow 27x = 243 \rightarrow x = 9$ overtime hours.
16. (L1) $0.20 \times 500 = 100$ mL.
17. (L2) Salt stays $0.40(3) = 1.2$ L. $1.2 = 0.30(3 + w) \rightarrow 1.2 = 0.9 + 0.3w \rightarrow w = 1$ L.
18. (L2) $6x + 10(8 - x) = 7.50(8) \rightarrow -4x + 80 = 60 \rightarrow x = 5$ kg of the \$6 nuts.
19. (L3) $(10 + a)/(40 + a) = 0.40 \rightarrow 10 + a = 16 + 0.4a \rightarrow 0.6a = 6 \rightarrow a = 10$ mL.
20. (L1) $d = 60 \times 2 = 120$ km.
21. (L2) $21 \div 1.75 = 12$ km/h.
22. (L2) $t = 1500 \div 500 = 3$ hours.
23. (L3) Combined rate 160 km/h; $160t = 480 \rightarrow t = 3$ hours.
24. (L1) $0.15 \times 60 = \$9$.
25. (L2) $45 - 0.20(45) = 45 - 9 = \36 .
26. (L2) $1.13p = 56.50 \rightarrow p = \50.00 .
27. (L3) $1.20p = 3600 \rightarrow p = 3000$.
28. (L1) $4s = 36 \rightarrow s = 9$ cm.

- 29.** (L2) $x + 2x = 90 \rightarrow 3x = 90 \rightarrow x = 30$; angles 30° and 60° .
- 30.** (L2) $2(w + 3w) = 64 \rightarrow 8w = 64 \rightarrow w = 8$; width 8 m, length 24 m.
- 31.** (L3) $x + 2x + 3x = 180 \rightarrow 6x = 180 \rightarrow x = 30$; angles 30° , 60° , 90° .